SUPPLY CHAIN MANAGEMENT

Designing an End-to-End Supply Chain for

Tata Curvv EV

in India



Learning Team-2 | PGDM – SCM |

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01 Introduction & EV Market Context

India's EV Revolution

Tata Motors commands 66% of India's passenger EV market as of 2025 — the dominant leader in a fast-growing segment.

Tata Curvv EV

Launched August 2024 as a midsize electric SUV-coupé. Priced ₹17.49L–₹21.99L with a 502 km ARAI range.

Report Scope

End-to-end SCM framework analysis: strategy, network, drivers, risk, reverse logistics & performance metrics.

66%
EV Market Share (2025)

250K+
Cumulative EVs Sold

50K+
Curvv Units in 16 Months

4,300
Avg Monthly Sales

502km
ARAI Range 55 kWh var.

₹17.49L
Starting Price

02 Company Overview & EV Leadership

Founded 1945

Part of $165B Tata Group. India's largest auto manufacturer.

66% EV Market Share

250,000+ EVs sold. 2 lakh+ charging points & 100+ Mega Hubs.

Pune Assembly Plant

930-acre Pimpri-Chinchwad facility. 18 hrs cycle time, 150K units/yr capacity.

Agratas Gigafactory

$1.5B investment, Sanand Gujarat. Domestic cells from 2026, full cap by 2028.



03 Product & Customer Profile

Key Specifications

Parameter	Value
Battery	45 kWh (MR) / 55 kWh (LR)
ARAI Range	430 km / 502 km
Power	148 bhp / 165 bhp
0-100 km/h	8.6 seconds
DC Fast Charge	40 min (10–80%, 70kW)
Price	₹17.49L – ₹21.99L

Customer Profile

Age & Income

28–42 yrs, Urban Pro, HH ₹15–30L/yr

Mindset

Tech-savvy early adopters, eco-conscious, value OTA & connected features

Geography

Tier-1 cities = 70% sales

Competitors

Hyundai Creta EV, Mahindra BE6, MG ZS EV | Segment: ₹17–25L

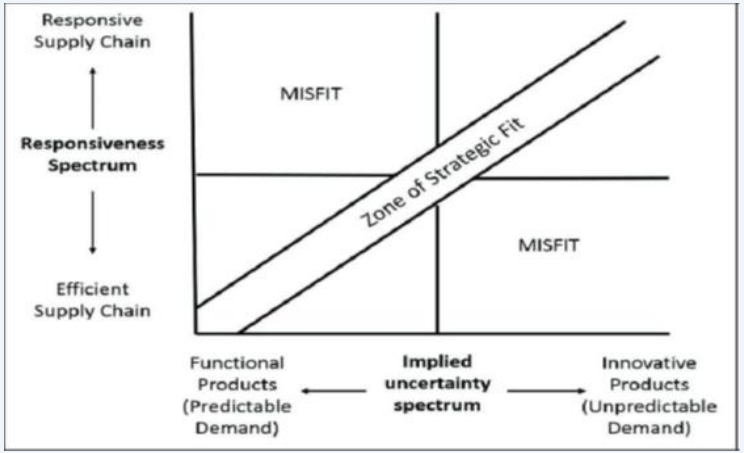
04 Supply Chain Strategy – Responsive-Efficient Hybrid

60% EFFICIENT

- Platform sharing with Nexon EV → 40% fewer components
- Vertical integration: battery packs, motors, body panels
- Build-to-order: 3–4 week lead time, minimal FG inventory
- 50,000+ units/yr drives supplier price negotiations

40% RESPONSIVE

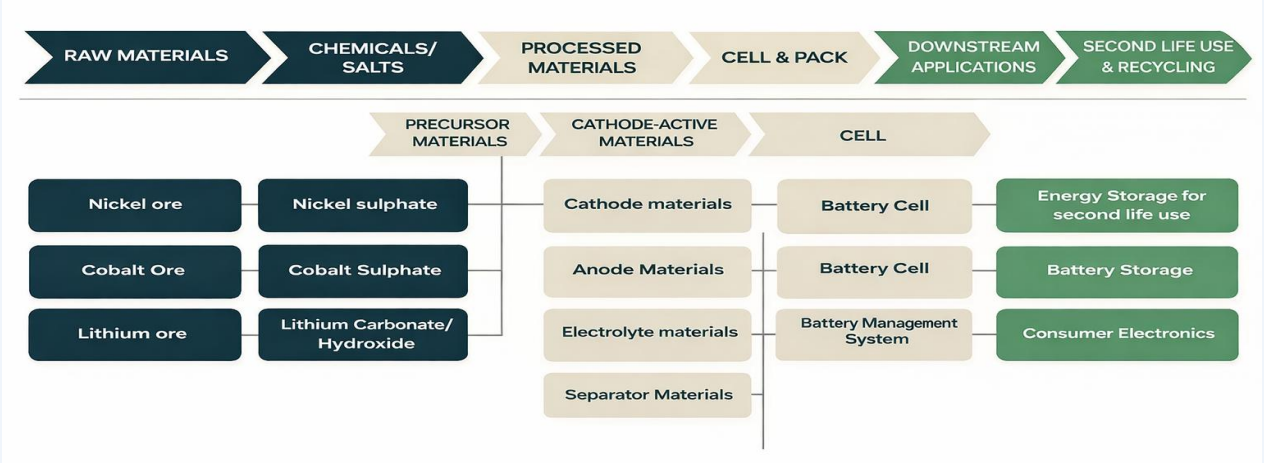
Chopra & Meindl – Strategic Fit



- Dual battery sourcing (Octillion + Tata AutoComp)
- Postponement – final config on firm orders only
- OTA updates – no physical recalls needed

05 Network Design & Battery Value Chain

EV Battery Value Chain



Curvv EV – 4-Tier Supply Network

TIER 3 Raw Materials	TIER 2 Components	TIER 1 Assembly	DISTRIBUTION
Lithium (Arg/Aus) Cobalt/Ni (Indonesia) Rare Earth (China) Steel – Tata Steel	Cells: Gotion/EVE Packs: Octillion/ Tata AutoComp Motor: Tata Sanand	Pune Plant 150K units/yr cap 35% utilization 18 hrs cycle time	8 Regional Hubs 150+ Dealerships 400+ Service Ctrs 2L+ Charge Points

06 Supply Chain Drivers – Chopra & Meindl Framework

Facilities

Modular lines
90K–150K units/yr

Trade-off: 18% cost saving at 80% util. vs. flexibility at 35%

Inventory

45-day battery buffer
₹450Cr (~15% COGS)

Trade-off: ₹75Cr extra WC vs. ₹180Cr disruption (ROI: 140%)

Transportation

70% road / 30% rail
5-day avg delivery

Trade-off: ₹8,500/veh vs ₹6,200 for speed & quality

Information

SAP S/4HANA + 1,500+ IoT
Forecast: 65%→82%

Trade-off: 22% safety stock cut 98% fill rate achieved

Sourcing

Multi-source, 70% local by 2028
Agratas cells 2026

Trade-off: Import risk now → 20-25% cost saving post-2026

Pricing

₹17.49L penetration price
FAME-II: ₹1.5L subsidy

Trade-off: 22% GM vs 28% ICE → 25%+ at 100K units

07 Risk Analysis & Coordination Plan

Risk	Prob	Impact	Priority	Mitigation Strategy
Battery Cell Supply Disruption	**7/10**	**9/10**	1	Dual sourcing + 45-day inventory + Agratas domestic production (2026)
Lithium Price Volatility	**8/10**	**8/10**	2	Long-term contracts (30%) + financial hedging (40% exposure)
Rare Earth Supply – China	6/10	9/10	3	Source diversification + R&D for alternative rare earth materials
Charging Infrastructure Gaps	7/10	6/10	4	Tata Power partnership + 2 lakh+ charging point expansion
Demand Volatility	8/10	7/10	5	Flexible capacity swing (90K–150K) + postponement strategy
Battery Degradation Quality	5/10	8/10	6	Multi-sourcing + rigorous cell-level testing protocols

Coordination: S&OP (Monthly) • CPFR with battery suppliers • VMI (40% components) • QBRs with 15 strategic Tier-1 partners • 18-month IBP rolling forecast

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08 Reverse Logistics & Sustainability

Battery End-of-Life Hierarchy

REFURBISH
SOH > 70%

Secondary automotive use
Module repair at Pune
5,000 packs/yr capacity

REPURPOSE
SOH 50–70%

2nd-life BESS with Tata Power
30% of retired batteries by 2028
Solar, grid balancing apps

RECYCLE
SOH < 50%

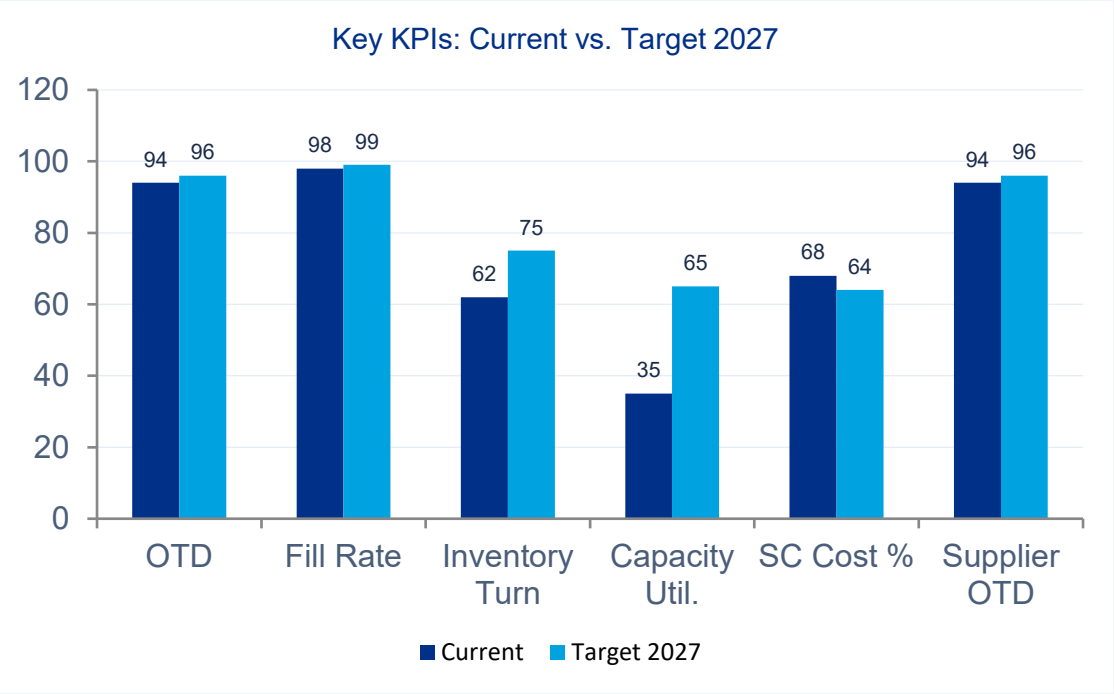
Attero, Lohum, Battery Smart
95% Li/Co/Ni recovery
Closed loop → Agratas

Circular Supply Chain Framework



18t CO_2e lifecycle • 12%→80% battery recycling (2030) • 35%→60% renewable energy (2027) • 2L+ charging points

09 KPI Dashboard & Key Trade-offs



24 days
Order→Delivery

2.8%
Warranty Rate

NPS 62
Net Promoter

97.5%
Inventory Accuracy

Key Trade-offs

Inventory vs Responsiveness

Cost: +₹75Cr WC

Gain: ₹180Cr disruption shield (ROI: 140%)

Flexibility vs Unit Cost

Cost: +12% fixed cost

Gain: Scale 50K→90K in 6 months

Localization vs Technology

Cost: 25% import premium

Gain: 502km range; 20-25% drop post-2026

Road vs Rail Transport

Cost: +₹2,300/vehicle

Gain: 48hr delivery, +15% conversion

10 Conclusion & Future Directions

Strategic Coherence

Hybrid 60-40 efficient-responsive strategy superior to pure-play approaches of rivals.

Network Optimization

Pune cluster + 8-hub distribution + Agratas $1.5B gigafactory = decisive vertical integration.

Risk Management

Battery supply mitigated via dual sourcing, 45-day buffer & domestic cell transition from 2026.

Circular Economy

400+ collection points, refurbish-repurpose-recycle hierarchy. Target 80% recycling by 2030.